

Exercise

1. 7 basic principles of object orientation

- **Divide and conquer:** break a problem into smaller, manageable parts by defining objects that interact with each other.
- **Encapsulation & modularity:** group related attributes and behaviors into objects, ensuring each object is self-contained.
- **Public interface:** clearly define how objects interact with each other using well-structured interfaces.
- **Information hiding:** hide internal details of an object, exposing only what is necessary.
- **Generality:** design objects for a general kind of task rather than a specific task, promoting reusability.
- **Extensibility:** allow objects to be extended to include new functionalities without modifying existing code.
- **Abstraction:** focus on essential characteristics of an object while ignoring irrelevant details.

2. benefits of OOP

- Save development time and cost by reusing code.
- Easier debugging as classes can be tested independently.
- Reused objects are already tested, reducing potential errors in new applications.